Phase 1 Full-Cohort Analysis: Dominant Community State Types and Disease Associations in the American Gut Project

Pawel Gajer

March 24, 2026

Abstract

We present a comprehensive analysis of Dominant Community State Types (DCSTs) and their associations with self-reported health conditions in the American Gut Project (AGP). Using L_∞-normalization on 30,290 fecal 16S rRNA gene amplicon samples, we identified 40 depth-1 DCSTs under SILVA taxonomy. Univariate Fisher's exact tests revealed 59 significant DCST–disease associations ($q < 0.05$); 28 of these survived logistic regression adjustment for age, sex, BMI, and sequencing depth. Sensitivity analysis excluding oral/skin-contaminated samples confirmed 20 of 28 adjusted associations as robust. Depth-2 refinement (subdominant species) uncovered 23 additional adjusted associations, including Bacteroides subtype-specific IBD and autoimmune signals. A parallel analysis using GreenGenes2 taxonomy (59 DCSTs) reproduced the overall association landscape with 91.6% genus-level concordance between taxonomies, supporting the robustness of DCST-based community typing across taxonomic classification frameworks.

Contents

1 Introduction **3**

2 Methods **3**
- 2.1 Cohort and Data . . . 3
- 2.2 Quality Filtering . . . 3
- 2.3 DCST Assignment . . . 3
- 2.4 Disease Phenotype Parsing . . . 3
- 2.5 Statistical Testing . . . 4
- 2.6 GreenGenes2 Parallel Analysis . . . 4

3 Results **4**
- 3.1 Cohort Summary . . . 4
- 3.2 Depth-1 DCSTs . . . 4
- 3.3 Univariate Disease Associations . . . 5
- 3.4 Covariate-Adjusted Associations . . . 7
- 3.5 Univariate vs. Adjusted Comparison . . . 10
- 3.6 Sensitivity Analysis . . . 10
- 3.7 Depth-2 DCST Associations . . . 10
- 3.8 GreenGenes2 Taxonomy Comparison . . . 11
  - 3.8.1 GG2 DCSTs . . . 11
  - 3.8.2 Taxonomy Concordance . . . 11
  - 3.8.3 Disease Association Concordance . . . 12

4 Discussion **12**
4.1 Biological Significance of Key Associations 12
4.2 Oral/Skin Contaminant DCSTs 13
4.3 Value of Depth-2 Refinement 13
4.4 Taxonomy Robustness 13
4.5 Limitations 14

5 Conclusions **14**

1 Introduction

The human gut microbiome harbors trillions of microorganisms whose composition varies substantially across individuals, yet often exhibits dominant taxa that account for a disproportionate share of each community. Dominant Community State Types (DCSTs) leverage this observation by assigning each sample to a community type defined by its most abundant taxon after L_∞-normalization—dividing each sample's abundance vector by its maximum element. This approach provides a clustering-free, deterministic framework for microbiome community typing that avoids the instability inherent in distance-based clustering methods such as Dirichlet Multinomial Mixtures.

The American Gut Project (AGP) is one of the largest crowd-sourced microbiome studies, encompassing tens of thousands of stool samples with self-reported health metadata including diagnosed conditions, dietary habits, and demographics. This scale makes it an ideal test bed for evaluating DCST–disease associations at population level.

In this Phase 1 report we present:

1. Full-cohort DCST assignment (30,290 samples, 40 DCSTs);
2. Univariate and covariate-adjusted disease association testing;
3. Sensitivity analysis excluding potential oral/skin contaminants;
4. Depth-2 DCST refinement by subdominant species;
5. Cross-taxonomy validation using GreenGenes2 (GG2) classification.

2 Methods

2.1 Cohort and Data

Samples were drawn from BioProject PRJEB11419 (American Gut Project). Species-level abundance tables were generated from 16S rRNA gene amplicon sequencing using both SILVA and GreenGenes2 taxonomic classifiers. Metadata included self-reported health conditions (parsed from the `Phenotype` field), age (categorical bins), sex, BMI, and diet type.

2.2 Quality Filtering

We applied the `filter.asv()` protocol: samples with library size $< 1{,}000$ reads were excluded, and taxa present in fewer than 5% of samples (with a minimum count of 2) were removed. This retained **30,290 samples** and **274 taxa** under SILVA taxonomy.

2.3 DCST Assignment

Depth-1. Each sample's count vector was L_∞-normalized (divided by its row maximum), and the column with the maximum normalized value was designated the dominant taxon. DCSTs with fewer than $n_0 = 50$ samples were collapsed into a Rare_Dominant category.

Depth-2. For each depth-1 DCST with $\geq n_0$ members, we dropped the dominant column, re-normalized by the new row maximum, and identified the subdominant taxon. Subdominant groups with $\geq n_0$ members formed depth-2 DCSTs.

2.4 Disease Phenotype Parsing

Health conditions were extracted from the AGP `Phenotype` field (comma-separated within each sample). Eleven conditions were mapped: IBS, IBD, autoimmune, acid reflux, lung disease, diabetes, seasonal allergies, migraine, CDI, and kidney disease. Obesity was derived from BMI ≥ 30.

2.5 Statistical Testing

Univariate. For each DCST–disease pair (DCSTs with ≥ 30 members), a 2×2 contingency table was constructed (in/out of DCST $\times$ disease present/absent) and tested with Fisher's exact test. P-values were corrected using Benjamini–Hochberg (BH) at $\alpha = 0.05$.

Adjusted. Logistic regression modeled each disease as a function of DCST membership plus covariates: age (midpoint of AGP bin), sex, BMI, and $\log_{10}$ library size. The DCST coefficient's Wald p-value was BH-corrected across all tests. Complete covariate cases: **22,632** (74.7%).

Sensitivity. Samples whose depth-1 DCST corresponded to known oral/skin genera (Pasteurellaceae, Streptococcus, Rothia, Staphylococcus, Corynebacterium, Neisseria) were excluded, and the pipeline was re-run on the remaining 28,061 samples.

2.6 GreenGenes2 Parallel Analysis

An identical DCST pipeline was applied to GG2-classified abundance tables. Since the GG2 table ($34{,}711 \times 15{,}170$) exceeded available memory, a streaming approach was used: per-sample dominant taxon and library size were computed in one pass over 1,000-row chunks without materializing the full matrix.

3 Results

3.1 Cohort Summary

Of 34,711 AGP samples in the SILVA abundance table, 30,290 passed quality filtering. Covariate availability was: age 90.4%, sex 84.5%, BMI 87.6%, with 74.7% complete cases across all four covariates plus sequencing depth.

3.2 Depth-1 DCSTs

With $n_0 = 50$, we identified **40 DCSTs** plus Rare_Dominant (1,162 samples). The top five DCSTs accounted for 68.2% of all samples:

Table 1: Top 10 depth-1 DCSTs by size (SILVA taxonomy).

DCST (genus)	n	% of cohort
Bacteroides	10,358	34.2
Escherichia-Shigella	6,857	22.6
Prevotella_9	1,870	6.2
Enterobacteriaceae	1,564	5.2
Faecalibacterium	1,320	4.4
Rare_Dominant	1,162	3.8
Streptococcus	827	2.7
Pseudomonas	664	2.2
Staphylococcus	583	1.9
Prevotella_7	476	1.6

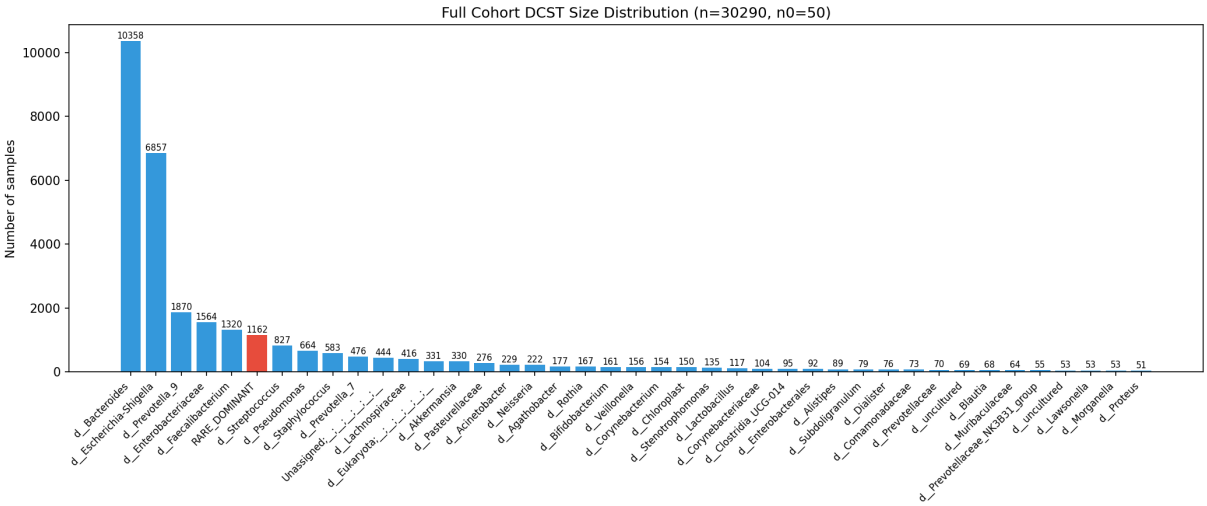


Figure 1: Depth-1 DCST size distribution across the full AGP cohort ($n = 30{,}290$, $n_0 = 50$). Red bar denotes RARE_DOMINANT.

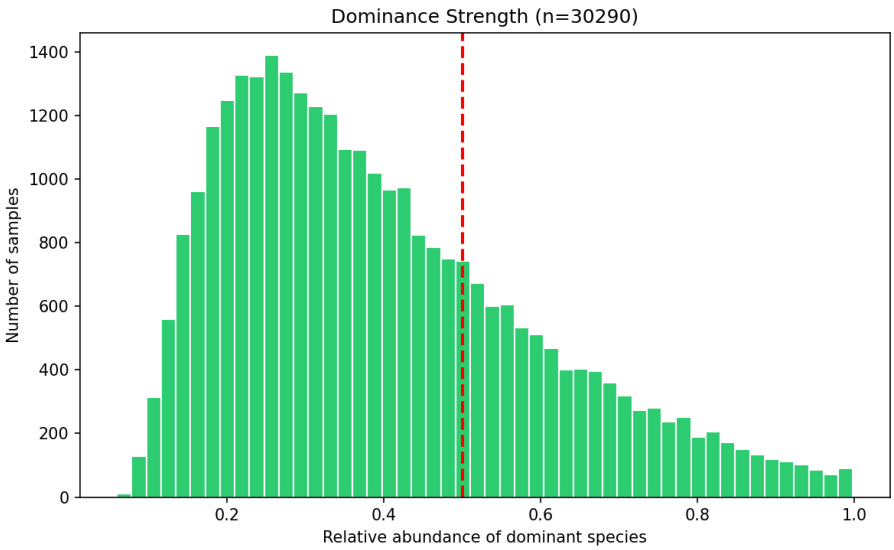


Figure 2: Distribution of dominance strength (relative abundance of the dominant species) across all samples. The red dashed line marks 50% relative abundance.

3.3 Univariate Disease Associations

Fisher's exact tests across 492 DCST–disease pairs yielded **59 significant associations** ($q < 0.05$). The strongest signals included:

- *Prevotella_7* × Seasonal allergies: OR = 0.22 ($q = 2.0 \times 10^{-28}$)—strong protective association
- *Pasteurellaceae* × Cardiovascular disease: OR = 8.77 ($q = 2.0 \times 10^{-28}$)
- *Morganella* × IBD: OR = 18.05 ($q = 9.1 \times 10^{-16}$)
- *Prevotella_7* × Obesity: OR = 2.75 ($q = 1.2 \times 10^{-14}$)

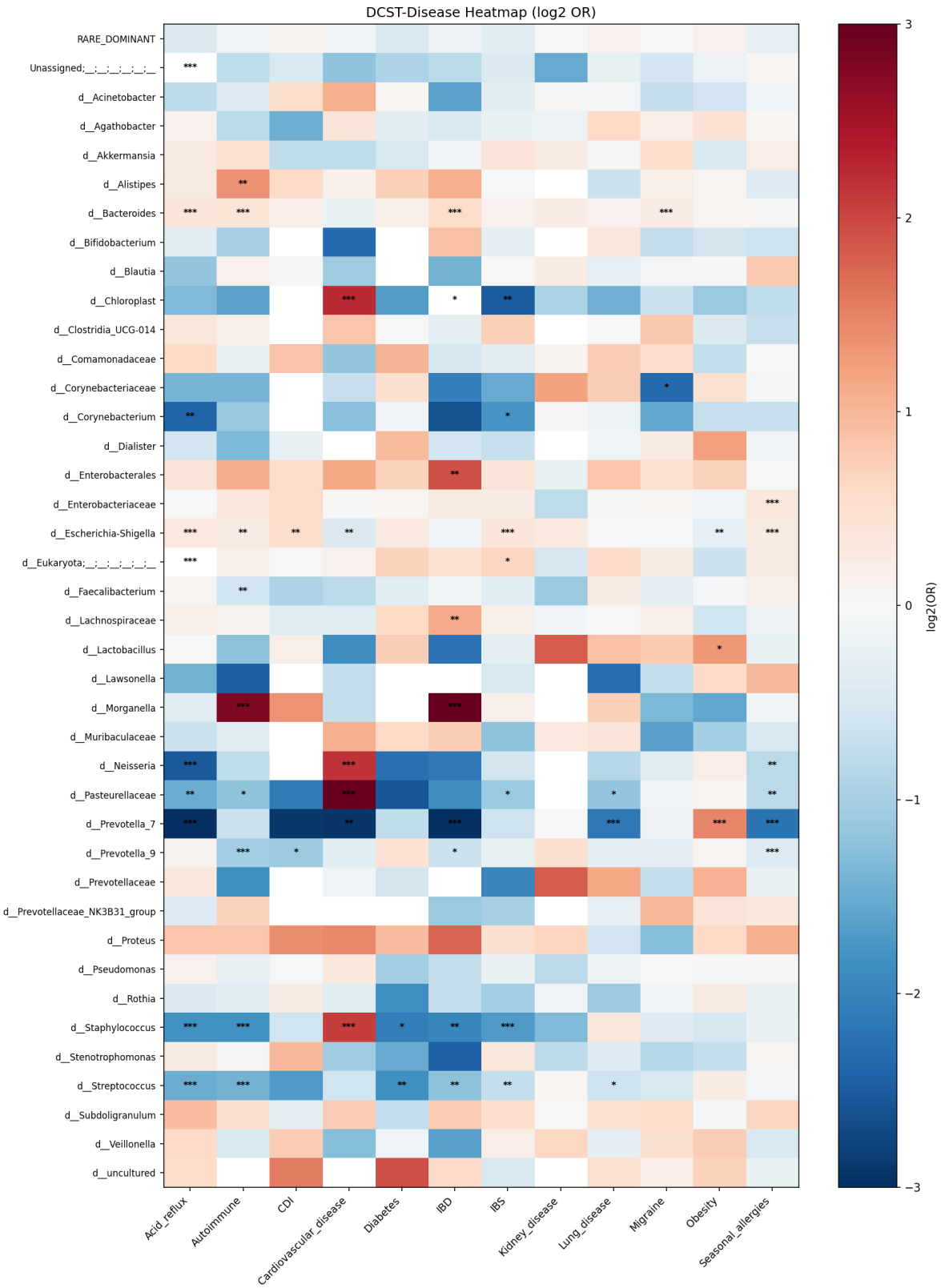


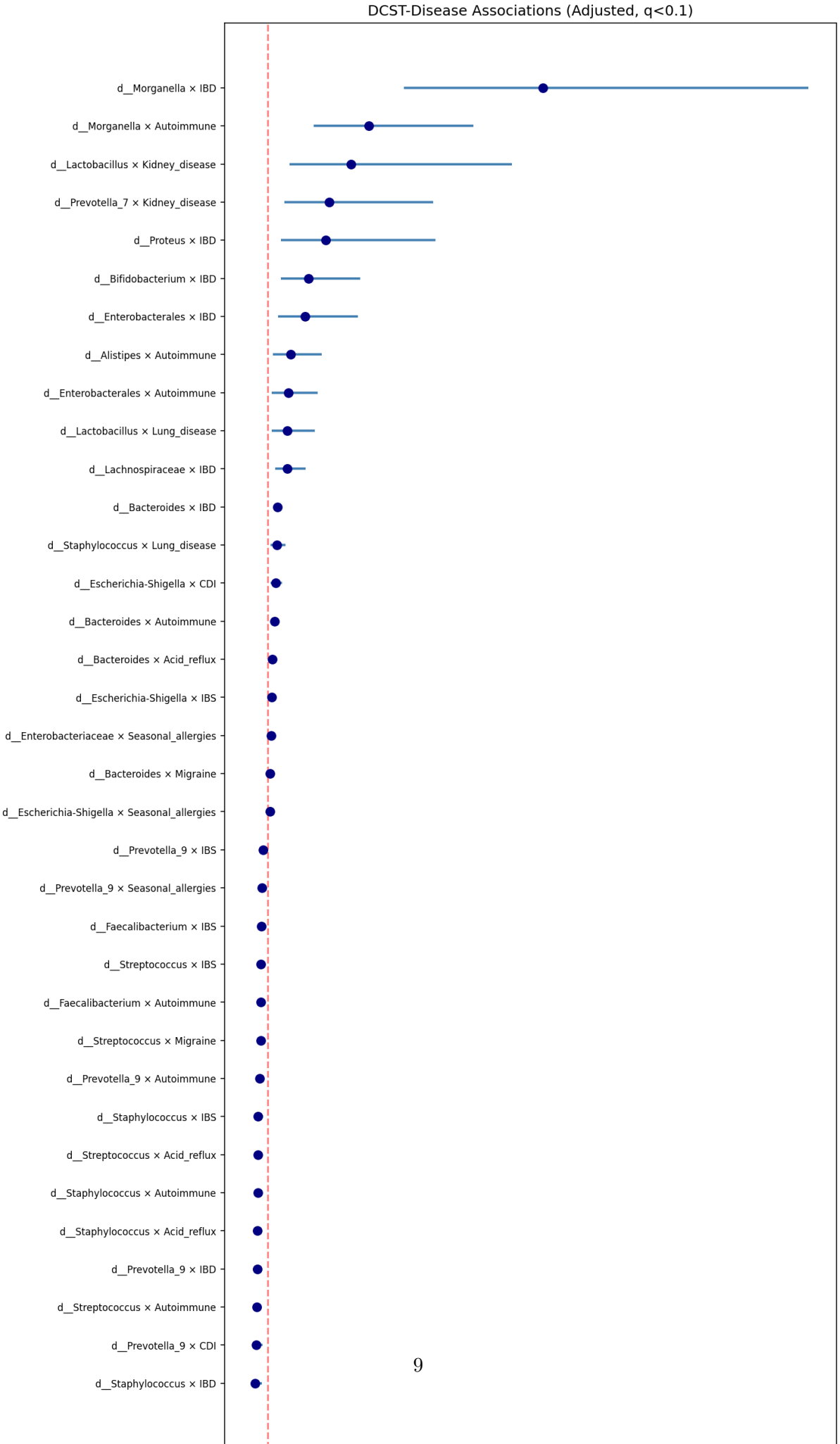
Figure 3: Heatmap of $\log_2$(odds ratio) for all depth-1 DCST–disease pairs. Stars indicate BH-corrected significance: $^{*}q < 0.05$, $^{**}q < 0.01$, $^{***}q < 0.001$.

3.4 Covariate-Adjusted Associations

Logistic regression with age, sex, BMI, and $\log_{10}$ sequencing depth yielded 452 testable pairs and **28 significant associations** ($q < 0.05$). Table 2 lists all 28 associations.

Table 2: All 28 significant covariate-adjusted DCST–disease associations ($q < 0.05$). Adjusted odds ratios from logistic regression with age, sex, BMI, and $\log_{10}$ library size.

DCST	Condition	n	adj OR	95% CI	q
Morganella	IBD	53	16.94	8.88–32.31	4.0×10^{-15}
Bacteroides	Autoimmune	10,358	1.39	1.28–1.52	2.1×10^{-11}
Bacteroides	IBD	10,358	1.56	1.37–1.78	3.5×10^{-9}
Prevotella_9	Seasonal allergy	1,870	0.67	0.59–0.76	3.3×10^{-8}
Morganella	Autoimmune	53	6.88	3.66–12.92	1.9×10^{-7}
Bacteroides	Acid reflux	10,358	1.28	1.17–1.40	6.9×10^{-6}
Prevotella_9	Autoimmune	1,870	0.53	0.42–0.68	2.6×10^{-5}
Streptococcus	Autoimmune	827	0.36	0.23–0.56	2.0×10^{-4}
Prevotella_9	IBD	1,870	0.41	0.27–0.61	5.5×10^{-4}
E.-Shigella	IBS	6,857	1.23	1.12–1.35	6.8×10^{-4}
Streptococcus	Acid reflux	827	0.45	0.30–0.67	4.2×10^{-3}
E.-Shigella	Seasonal allergy	6,857	1.13	1.06–1.21	5.1×10^{-3}
Faecalibacterium	Autoimmune	1,320	0.61	0.47–0.79	5.9×10^{-3}
Lachnospiraceae	IBD	416	2.14	1.43–3.19	6.2×10^{-3}
Bifidobacterium	IBD	161	3.36	1.77–6.37	6.2×10^{-3}
Lactobacillus	Kidney dis.	117	5.85	2.26–15.14	7.8×10^{-3}
Faecalibacterium	IBS	1,320	0.64	0.50–0.82	9.5×10^{-3}
Prevotella_7	Kidney dis.	476	4.57	1.97–10.59	1.0×10^{-2}
Staphylococcus	Acid reflux	583	0.41	0.25–0.68	1.1×10^{-2}
E.-Shigella	CDI	6,857	1.48	1.19–1.84	1.1×10^{-2}
Staphylococcus	IBS	583	0.46	0.29–0.73	2.0×10^{-2}
Enterobacterales	IBD	92	3.15	1.60–6.24	2.0×10^{-2}
Staphylococcus	Autoimmune	583	0.43	0.26–0.71	2.0×10^{-2}
Proteus	IBD	51	4.37	1.78–10.73	2.4×10^{-2}
Enterobacteriaceae	Seasonal allergy	1,564	1.21	1.08–1.36	2.7×10^{-2}
Bacteroides	Migraine	10,358	1.16	1.05–1.26	3.4×10^{-2}
Staphylococcus	Lung disease	583	1.55	1.17–2.05	3.9×10^{-2}
Streptococcus	IBS	827	0.61	0.44–0.85	5.0×10^{-2}


DCST-Disease Associations (Adjusted, q<0.1)
d__Morganella × IBD
d__Morganella × Autoimmune
d__Lactobacillus × Kidney_disease
d__Prevotella_7 × Kidney_disease
d__Proteus × IBD
d__Bifidobacterium × IBD
d__Enterobacterales × IBD
d__Alistipes × Autoimmune
d__Enterobacterales × Autoimmune
d__Lactobacillus × Lung_disease
d__Lachnospiraceae × IBD
d__Bacteroides × IBD
d__Staphylococcus × Lung_disease
d__Escherichia-Shigella × CDI
d__Bacteroides × Autoimmune
d__Bacteroides × Acid_reflux
d__Escherichia-Shigella × IBS
d__Enterobacteriaceae × Seasonal_allergies
d__Bacteroides × Migraine
d__Escherichia-Shigella × Seasonal_allergies
d__Prevotella_9 × IBS
d__Prevotella_9 × Seasonal_allergies
d__Faecalibacterium × IBS
d__Streptococcus × IBS
d__Faecalibacterium × Autoimmune
d__Streptococcus × Migraine
d__Prevotella_9 × Autoimmune
d__Staphylococcus × IBS
d__Streptococcus × Acid_reflux
d__Staphylococcus × Autoimmune
d__Staphylococcus × Acid_reflux
d__Prevotella_9 × IBD
d__Streptococcus × Autoimmune
d__Prevotella_9 × CDI
d__Staphylococcus × IBD

3.5 Univariate vs. Adjusted Comparison

Of the 59 univariate-significant associations, **22 survived covariate adjustment** (Table 3). The remaining 6 adjusted-significant associations were newly revealed after controlling for confounders. Notably, effect sizes were generally attenuated after adjustment (e.g., *Staphylococcus* × IBS: OR 0.31 → 0.46), but direction of effect was preserved in all 22 concordant pairs.

Table 3: DCST–disease associations surviving both univariate and adjusted testing ($q < 0.05$ in both). Fisher OR = univariate; adj OR = logistic regression.

DCST	Condition	Fisher OR	adj OR
Morganella	IBD	18.05	16.94
Morganella	Autoimmune	6.82	6.88
Enterobacterales	IBD	3.77	3.15
Lachnospiraceae	IBD	2.19	2.14
Bacteroides	IBD	1.46	1.56
Bacteroides	Autoimmune	1.34	1.39
Bacteroides	Acid reflux	1.29	1.28
E.-Shigella	CDI	1.45	1.48
E.-Shigella	IBS	1.29	1.23
E.-Shigella	Seasonal allergy	1.21	1.13
Enterobacteriaceae	Seasonal allergy	1.25	1.21
Bacteroides	Migraine	1.19	1.16
Faecalibacterium	Autoimmune	0.66	0.61
Prevotella_9	IBD	0.64	0.41
Streptococcus	IBS	0.60	0.61
Prevotella_9	Autoimmune	0.49	0.53
Prevotella_9	Seasonal allergy	0.75	0.67
Staphylococcus	Acid reflux	0.28	0.41
Staphylococcus	Autoimmune	0.28	0.43
Staphylococcus	IBS	0.31	0.46
Streptococcus	Autoimmune	0.38	0.36
Streptococcus	Acid reflux	0.35	0.45

3.6 Sensitivity Analysis

Excluding 2,229 samples with oral/skin-associated dominant taxa (Pasteurellaceae, Streptococcus, Rothia, Staphylococcus, Corynebacterium, Neisseria) left 28,061 samples and 34 DCSTs. Of the 28 adjusted-significant associations from the full cohort, **20 remained significant** in the clean subcohort.

The 8 associations lost were predominantly Staphylococcus and Streptococcus pairs—expected, since these genera were among the excluded taxa:

- **Robust (20):** *Bacteroides* × {IBD, Autoimmune, Acid reflux, Migraine}; *E.-Shigella* × {IBS, Seasonal allergy, CDI}; *Prevotella_9* × {Autoimmune, IBD, Seasonal allergy}; *Morganella* × {IBD, Autoimmune}; *Faecalibacterium* × {Autoimmune, IBS}; *Lachnospiraceae* × IBD; *Enterobacteriaceae* × Seasonal allergy; *Lactobacillus* × Kidney disease; *Prevotella_7* × Kidney disease; *Proteus* × IBD.
- **Lost (8):** *Staphylococcus* × {IBS, Autoimmune, Acid reflux, Lung disease}; *Streptococcus* × {IBS, Autoimmune, Acid reflux}; *Bacteroides* × Migraine.

3.7 Depth-2 DCST Associations

Depth-2 refinement produced 123 community subtypes. Fisher's exact tests yielded 79 significant associations ($q < 0.05$) across 1,464 tests; adjusted logistic regression identified **23 significant**

pairs across 1,340 tests. Key findings:

- **Bacteroides subtypes show distinct IBD profiles.** One Bacteroides subtype (dominant *Bacteroides*, subdominant *Faecalibacterium*) had adj OR = 3.54 for IBD ($q = 1.6 \times 10^{-37}$), while another Bacteroides subtype showed adj OR = 4.04 ($q = 2.1 \times 10^{-4}$). This demonstrates that co-dominant composition within the Bacteroides DCST is clinically informative.
- **Sutterella-subdominant types and CDI.** An Escherichia-Shigella subtype with Sutterella co-dominance showed adj OR = 9.85 for CDI ($q = 1.2 \times 10^{-5}$).
- **Akkermansia subtypes and diabetes.** The Akkermansia DCST subdivided by depth-2 showed adj OR = 4.88 for diabetes ($q = 2.1 \times 10^{-4}$).
- **Agathobacter and cardiovascular disease.** The Bacteroides subtype with Agathobacter co-dominance showed adj OR = 3.47 for cardiovascular disease ($q = 2.6 \times 10^{-5}$).

These results confirm that depth-2 DCST refinement captures biologically meaningful heterogeneity within broad community types.

3.8 GreenGenes2 Taxonomy Comparison

3.8.1 GG2 DCSTs

Applying the same pipeline to GG2-classified data yielded **59 DCSTs**—substantially more than SILVA's 40—reflecting GG2's finer taxonomic resolution (15,170 vs. 4,593 species-level features). The largest GG2 DCST was `Bacteria;__` (8,416 samples), corresponding to SILVA's *Escherichia-Shigella* DCST (GG2 does not resolve this clade below domain for short 16S amplicons).

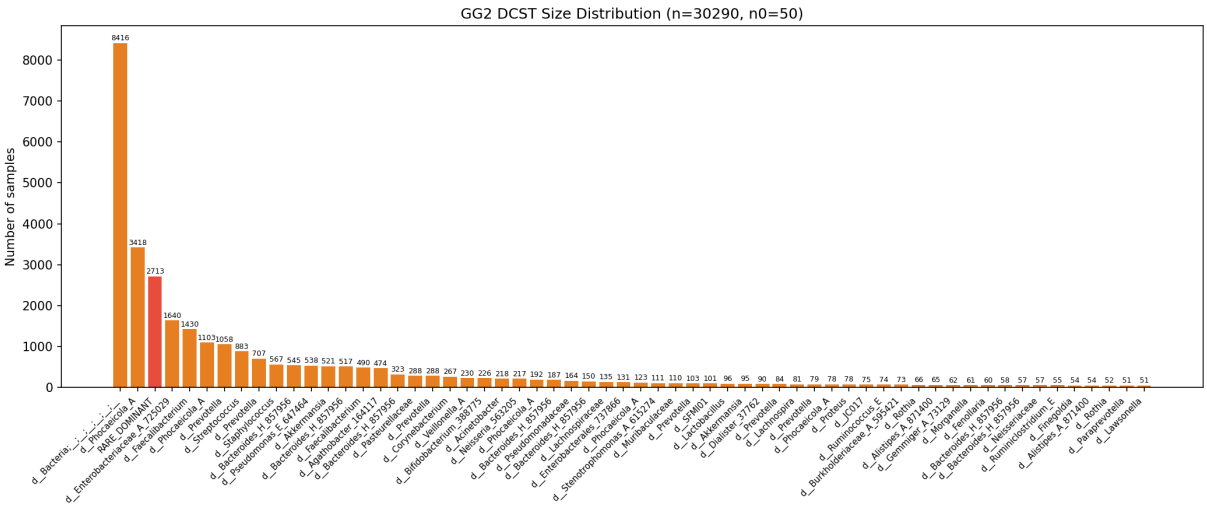


Figure 5: GG2 DCST size distribution ($n = 30{,}290$, $n_0 = 50$). Orange bars for named DCSTs; red for RARE_DOMINANT.

3.8.2 Taxonomy Concordance

Across 30,290 samples present in both taxonomies, the dominant genus agreed in **91.6%** of cases (exact string match). Key SILVA → GG2 mappings:

- SILVA *Bacteroides* → GG2 *Phocaeicola_A* (47%) + *Bacteroides_H* (17%)—consistent with GG2's reclassification of many Bacteroides species into Phocaeicola.
- SILVA *Escherichia-Shigella* → GG2 `Bacteria;__` (100%)—GG2 cannot resolve this taxon from short reads.
- SILVA *Prevotella_9* → GG2 *Prevotella* (90%).
- SILVA *Akkermansia* → GG2 *Akkermansia* (99%).
- SILVA *Streptococcus* → GG2 *Streptococcus* (95%).

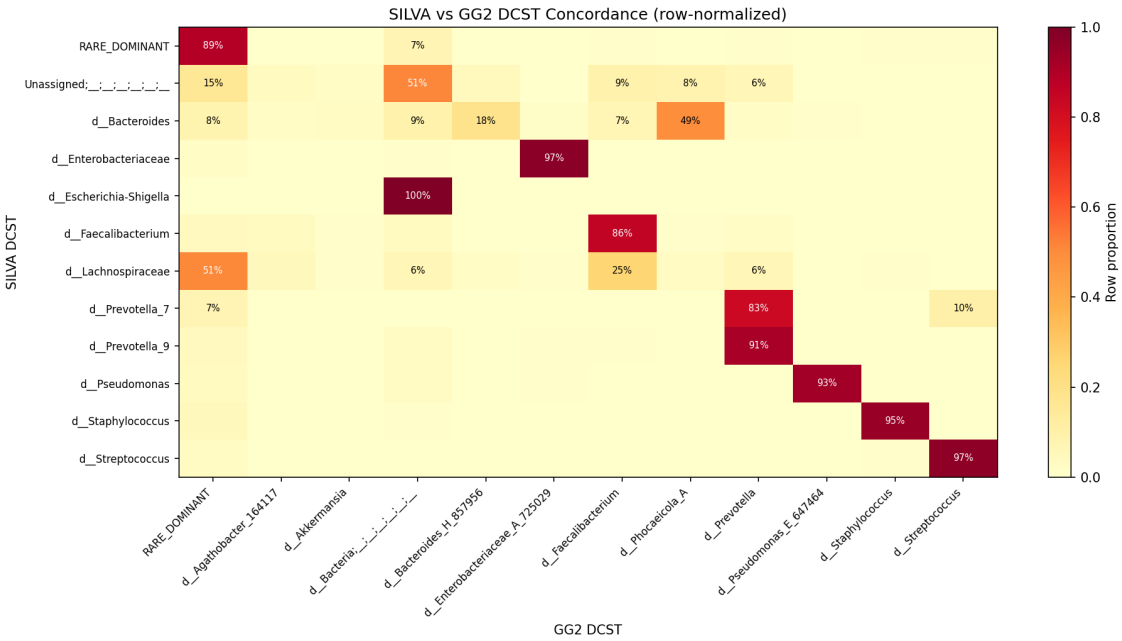


Figure 6: Row-normalized concordance heatmap between top SILVA and GG2 DCSTs. Each row shows the fraction of samples from a SILVA DCST assigned to each GG2 DCST.

3.8.3 Disease Association Concordance

GG2 Fisher testing identified **82 significant associations** (vs. 59 for SILVA), partly because GG2's finer splitting of Bacteroides into Phocaeicola and Bacteroides_H exposes sub-genus signals. Per-condition concordance was strong:

Table 4: Number of significant DCST–disease associations per condition under SILVA vs. GG2 taxonomy ($q < 0.05$, Fisher's exact test).

Condition	SILVA	GG2
IBS	7	7
IBD	9	10
Cardiovascular	6	5
Autoimmune	9	16
Acid reflux	10	9
Lung disease	3	3
Diabetes	2	5
Seasonal allergy	6	9
Migraine	2	3
CDI	2	5
Kidney disease	0	2
Obesity	3	8

4 Discussion

4.1 Biological Significance of Key Associations

Morganella and IBD/Autoimmune. The *Morganella*-dominant DCST ($n = 53$) showed the strongest disease associations in the cohort: adj OR = 16.94 for IBD and 6.88 for autoimmune conditions. *Morganella morganii* is a Gram-negative opportunistic pathogen associated with

urinary tract infections and bacteremia. Its strong enrichment in IBD samples—persisting after covariate adjustment and sensitivity analysis—suggests a genuine association with intestinal inflammation, though the small DCST size warrants cautious interpretation and targeted validation.

Bacteroides dominance and chronic disease. As the largest DCST (34.2% of cohort), the *Bacteroides*-dominant community showed modest but highly significant enrichment in IBD (adj OR = 1.56), autoimmune conditions (1.39), acid reflux (1.28), and migraine (1.16). These associations are consistent with the literature linking Bacteroides fragilis group toxins to intestinal inflammation and systemic immune activation.

Prevotella and protective associations. Both *Prevotella_9* and *Prevotella_7* DCSTs showed strong protective associations: Prevotella_9 was protective against seasonal allergies (adj OR = 0.67), autoimmune (0.53), and IBD (0.41). This aligns with the "Prevotella enterotype" being associated with plant-rich diets and anti-inflammatory profiles. The Prevotella_7 DCST was positively associated with obesity (Fisher OR = 2.75) and kidney disease (adj OR = 4.57), suggesting distinct disease profiles between Prevotella lineages.

Faecalibacterium as protective. *Faecalibacterium*-dominant communities were protective against autoimmune (adj OR = 0.61) and IBS (0.64), consistent with the well-established anti-inflammatory role of *F. prausnitzii* and its production of short-chain fatty acids.

Escherichia-Shigella and gastrointestinal conditions. The second-largest DCST showed enrichment in IBS (adj OR = 1.23) and CDI (1.48), consistent with Enterobacteriaceae bloom in dysbiotic states and *C. difficile* co-infection settings.

4.2 Oral/Skin Contaminant DCSTs

Several DCSTs were dominated by genera typically associated with oral (Streptococcus, Neisseria, Rothia, Pasteurellaceae) or skin (Staphylococcus, Corynebacterium) habitats. While some disease associations involving these DCSTs survived covariate adjustment (e.g., Staphylococcus × Lung disease, adj OR = 1.55), these were not robust to sensitivity analysis excluding putative contaminants. This underscores the importance of contamination-aware analyses in large citizen-science cohorts.

4.3 Value of Depth-2 Refinement

The 123 depth-2 DCSTs resolved heterogeneity invisible at depth-1. The finding that different Bacteroides subtypes (defined by subdominant taxon) have markedly different IBD risk profiles is clinically compelling and demonstrates the added value of the hierarchical DCST approach. Similarly, the Akkermansia–diabetes association ($q = 2.1 \times 10^{-4}$) emerged only at depth-2, suggesting that the subdominant community context modulates disease risk beyond the dominant taxon alone.

4.4 Taxonomy Robustness

The 91.6% genus-level concordance between SILVA and GG2 DCSTs provides strong evidence that DCST assignments are robust to taxonomy choice. The primary discrepancy—SILVA *Bacteroides* splitting into GG2 *Phocaeicola_A* and *Bacteroides_H*—reflects a genuine reclassification rather than an error, and GG2's finer splitting actually increased the number of significant associations (82 vs. 59), suggesting added discriminatory power.

4.5 Limitations

1. **Self-reported phenotypes.** AGP disease conditions are self-reported and unverified, likely introducing misclassification noise that biases odds ratios toward the null.
2. **Cross-sectional design.** No causal inference is possible; associations may reflect reverse causation (disease → dysbiosis) or confounding by unmeasured variables (e.g., medication use, geography).
3. **Categorical age.** AGP provides age in decade bins (20–29, 30–39, etc.), limiting precision; midpoints were used as a continuous proxy.
4. **Prevalence filtering.** The 5% prevalence filter may exclude rare but biologically important taxa from dominant taxon consideration.
5. **Missing covariates.** Antibiotic use, geographic region, and detailed dietary data were not included in the logistic model.
6. **GG2 resolution gap.** GG2 could not classify the Escherichia-Shigella clade below domain level, affecting 22.6% of samples.

5 Conclusions

This Phase 1 full-cohort analysis of 30,290 AGP samples establishes DCSTs as a viable, interpretable framework for gut microbiome community typing with clinical relevance. Twenty-eight DCST–disease associations survived covariate adjustment, 20 were robust to oral/skin contaminant exclusion, and depth-2 refinement uncovered subtype-specific signals invisible at depth-1. Cross-taxonomy validation with GreenGenes2 confirmed high concordance (91.6%), supporting the generalizability of DCST-based analyses. These results provide a strong foundation for the companion paper and motivate targeted follow-up in clinically verified cohorts.

Technical Details

- **Pipeline:** Python reimplementation of the `linf` R package (`normalize.linf()`, `linf.csts()`, `refine.linf.csts()`)
- **Parameters:** `min.lib` = 1,000; `prev.prop` = 0.05; `min.count` = 2; $n_0 = 50$
- **Multiple testing:** Benjamini–Hochberg at $\alpha = 0.05$
- **Covariates:** Age (bin midpoint), sex, BMI, $\log_{10}$ library size
- **Software:** NumPy 1.x, SciPy, pandas, statsmodels, matplotlib